

Valerie's notes

- ✓ In the title page, break the line of the title after 'Tipping points', so that 'early warning signals' are on the next line
- ✓ Page iii, line 9 from bottom - "true asymptotic power-law scaling"
- ✓ Page iv, item 2, I think you mean Livina et al, Climate Dynamics 2011; you also have to mention here the paper by Ashwin et al, Phil Trans Royal Soc A 2012 - say that this paper later expanded on the types of tipping (although we mentioned them first)
- ✓ Page iv, lines 3-1 from the bottom: replace with the shorter text: "The study in section 2.3.5 also shows that we can expect better performance of the PS indicator..." At the end of this sentence add the following: "We use the PS indicator not as a precise measure of scaling but as a sensor of the changes in the temporal organisation of the time series".
- ✓ Page v, lines 4-5 from top: delete the whole sentence "This is unfortunate.." Instead, put the following sentence: "Unfortunately, in many real systems gathered data may be either short/incomplete, or the dynamics of the process very fast compared with the sampling rate of measurement; even in these circumstances our techniques may be useful, as shown in the case studies".
- ✓ Page v, item 4: "One hypothesis of this thesis is that the PS-indicator is better suited for EWS analysis in this scenario than other indicators, and in section 2.3.7..."
- ✓ Page vi, 10th line from bottom: "...does not exhibit true power-law scaling in asymptotic."
- ✓ Page viii, 9th line from top: "This was an unfortunate issue..."
- ✓ Page viii, lines 13-14 from top: "We agree that the argument for the presence of EWS was based on ad hoc analysis, and although..."
- ✓ Page viii, lines 16-17 from top: "It is unfortunate that a system with more convincing results was not available (despite our attempts to obtain, for instance, land model data from German colleagues, which is still not in public domain). Other suggestions..."
- ✓ Page viii, fix typo "whoch". Run some spellcheck for latex through the thesis.
- ✓ Page ix, 7th line from top: "...this is, unfortunately, an unresolved issue."
- ✓ Page xvi, 5th line from bottom: "...of the time step used in the integration of the system, or the sampling rate of the measurements."
- ✓ Page xxviii, "codes" (plural)
- ✓ Page xxix-xxx, try to vary font or margins to put Glossary into one page (I would suggest a smaller font)
- ✓ Page 1, 9th line from bottom: replace "or simply vanishes" with "and the number of states changes".
- In Fig.1.5, the curve in the left panel is not black but grey - it is a minor issue, so you may leave it as it is
- ✓ Page 9, 5th line top, "both of length 10^6 data points"
- ✓ Page 9, 8th line from top: remove "also"
- ✓ Page 9, in red font "this higher-dimensionality" - I don't think that the dash is needed
- Page 9, the title of section 1.2 has no following text - use \newpage to move this title to the next page. Check at the end, after the final compilation, that all sections headers are properly placed and not separated from their text.
- ✓ Page 11, reference Livina should be in square brackets (not only the years of the references. Here, after reference [Livina et al 2011] add reference [Ashwin et al 2012]
- ✓ Page 27, line 2 from top: add backslash \ to make symbol theta

- ✓ Check your systems of equations: in some you use commas after first equation, in others you only put a dot of full stop at the end of the last equations. Choose one style and follow it in all cases.
- ✓ Page 29, at the end of section 1.4, remove brackets around "which is the actual moment of bifurcation", just separate it by comma.
- ✓ Page 30, line 14 from bottom: replace "to produce" with "to detect"
- ✓ Page 30, lines 14-13 from bottom: "...are applicable to a wide range of dynamical systems exhibiting a variety of tipping points...."
- ✓ Page 31, first sentence of section 1.6: remove "elsewhere"
- ✓ Page 34, 1st line: correct "assess"
- ✓ Page 34, the last line of section 1.6: delete word "clearly", as it is followed by "clear" in the same sentence.
- ✓ Page 39, in the beginning of section 2.1.1, you need to add the ACF-related reference [Held & Kleinen 2004]
- ✓ Page 40, line 5: remove double "since"
- ✓ Page 41, 15th line from bottom: "In practice,"
- ✓ Page 45, at the end of equation 2.15 there should be a comma (not dot)
- ✓ Page 47, 11th line from bottom - "assuming"
- ✓ Page 48, in the caption of Fig.2.3 put 'respectively' in full.
- ➡ Regarding the logarithmic binning, you may like to have a look at Fig.3 in the paper [Ashkenazy & Gildor, JGR 2009] - panel 2 shows the curve of the logarithmically binned PS; you may like to create a similar figure. In your figure 2.18c logarithmically binned PD is somehow much shorter than in the original range - maybe try another bin size?
- ✓ Not sure if you need all three equations in (2.26), as this is a straightforward derivation, you may leave only the last one; or put them in one line with logical arrow of derivation?
- ✓ Page 53, add a comma at the end of equation 2.39
- ✓ In some places 'non stationarity' has a dash, in others not; check for consistency
- ✓ In page 54, 3rd line from top, remove "simply", as it repeats in the next line
- ✓ In page 65, you don't need a comma in the first line of equation 2.65, as equality sign is assumed there; the same in equations 2.67 in page 66, 2.84 in page 74, 2.86 in page 75
- ✓ In page 69, put comma at the end of equation (2.71) and start 'where' in the next line with a small letter
- ✓ Do you need to number all lines in derivation 2.74-2.77? If you are using 'equation' Latex environment, put \nonumber in those lines where you don't need a number
- ✓ Page 71, in 4th line from bottom: "...there is no value..."
- ✓ Equations 2.65 (page 65) and 2.84 (page 74) are identical. I understand you need them for the flow of text; maybe second time give only second line and use the earlier equation number as a reference? Later in Eq. 3.44 you mention that it reprints 3.25, which may be a good style for all such cases
- ✓ Page 75, in lines 7-8 from top you explain what you mean by log-log, but in fact you use the idea much earlier. You may like to move the text in brackets to the first place where log-log is mentioned
- ✓ Page 95, the first sentence of section 2.5.1 uses several "uses" - try to reformulate it.
- ✓ Page 108, in the first text line under Fig.2.24, "...surprising that we see any trend..."
- ✓ Page 109, in the Discussion, you need to add a note on the robustness of the PS-indicator to periodic oscillations in the data
- ✓ Page 120, in the paragraph in red font, add a blank space after reference 3.3.3.

- ✓ Page 121, lines 1-2 from top: "The eigenvalues of the relevant estimated Jacobian, which are..."
- Depending on the new layout, you may need to use \newpage before section 3.4.1
- ✓ When discussing the results in Fig.12, you say that there are no EWS observed in any indicators. But I think ACF1 of EOF does show EWS, doesn't it?
- ✓ Page 171, in the first paragraph, again add reference to Ashwin et al 2012 after reference Livina et al. In the next sentence down, delete word 'Similarly' (it repeats the same in the previous sentence)
- ✓ Page 178, 2nd line from top, "...but it does not appear..."
- ✓ Page 178, last line, "...true power-law scaling in asymptotic."
- ✓ Page 183, line 4 from top, you are talking about "wrong lag", but since for the ACF-indicator one always uses the same lag 1, you Amy like to discuss the sampling rate of the time series. Obviously, by varying lag you can compensate for this effect. You may like to add some text on this.
- ✓ Pages 181, 197, 203: the presented results were published (to avoid saying "presented" twice)
- ✓ Page 205, 3rd line from top: "...hurricane, which gives..."
- ✓ Page 206, 8th line from bottom: "...indicators obtained in section..." (remove "which have been")
- ✓ Fig. 4.19 looks like low-resolution jpeg in its error bars. Check if you have eps-file to include
- ➡ Conclusions (pages 217-227) is not marked in red font, although I presume this is new text - mark it for the examiners?
- ✓ Page 217, lines 2-3 in first paragraph: delete "Although others have developed strategies along different lines,"
- ➡ Page 220, line 13 from bottom, remove double dot after 'etc'
- ✓ Page 222, the footnote should be a complete sentence, starting with a capital W and with a dot at the end. In the footnote in page 223, add dot, too.
- ✓ Page 224, 2nd paragraph from bottom, "In summary...."
- ✓ Page 225, lines 10-11 from top, "...a long-term research goal."
- ✓ Page 227, 6th line from bottom: "higher time resolution"
- ✓ Page 227, line 5 from bottom: "EWS methods" (remove "indicator")
- ✓ Page 239, four lines of expressions for d11..d22 are wider than line width - take this block into small font like {\small }
- ✓ The same about page 241
- ✓ When you say here that you were using {\it computer}, was that something Waterloo Maple or Matlab Symbolic Toolbox? If so, give a reference to the software.
- ✓ Page 243, for the pieces of code it is better to use \tt font